

# 1<sup>st</sup> order and 1<sup>st</sup> degree ODE

The most general first order and first-degree ODE is:  $\frac{dy}{dx} = f(x, y)$

The integration of this equation gives its general solution. Direct integration is not always possible. The methods of identification to make this ODE integrable are the objective of the rest of this chapter. The following procedures will help in this respect.

- Separation of variables
- Homogeneous equations
- Reducible to homogeneous equation
- Linear equations
- Reducible to Linear equations
- Exact equation

# Separation of variables

A first-order ODE is separable if it can be written in the form:

$$P(x, y) + Q(x, y) \frac{dy}{dx} = 0$$

$$\text{Or, } P(x, y) dx + Q(x, y) dy = 0$$

Where,  $P(x, y)$  and  $Q(x, y)$  are functions of  $x$  and  $y$ .

If these equations can be written as

$$f_1(x) + f_2(y) \frac{dy}{dx} = 0$$

$$\text{Or, } f_1(x) dx + f_2(y) dy = 0$$

by some algebraic manipulation, then the variables are said to be separated. Integrating on both sides,

$$\int f_1(x) dx + \int f_2(y) dy = c$$

$$\text{Or, } F_1(x) + F_2(y) = c$$

# Separation of variables

**Problem#1.** Solve the ODE:  $(x^2 + 1)(y^2 - 1) dx + x y dy = 0$

Solution: Given  $(x^2 + 1)(y^2 - 1) dx + x y dy = 0 \dots \dots (i)$

Dividing both sides of Eq. (i) by  $(y^2 - 1)x$ , we get

$$\frac{(x^2 + 1)}{x} dx + \frac{y}{y^2 - 1} dy = 0$$

$$\text{Integrating, } \int \frac{(x^2 + 1)}{x} dx + \int \frac{y}{y^2 - 1} dy = c_1$$

$$\text{or, } \int \left( x + \frac{1}{x} \right) dx + \int \frac{y}{y^2 - 1} \frac{d(y^2 - 1)}{2y} = c_1$$

$$\text{or, } \frac{x^2}{2} + \ln(x) + \frac{1}{2} \ln(y^2 - 1) = c_1$$

$$\text{Or, } \ln(y^2 - 1) = 2c_1 - x^2 - \ln x^2$$

$$\text{or, } y^2 - 1 = e^{2c_1 - x^2 - \ln x^2}$$

$$\therefore y^2 = 1 + \frac{1}{x^2} c e^{-x^2}$$

# Separation of variables

**Problem#2.** Solve the ODE  $\frac{d y}{d x} = e^{x+y} + x^2 e^{x^3+y}$

Solution: Given  $\frac{d y}{d x} = e^{x+y} + x^2 e^{x^3+y} \dots \dots \dots (i)$

Dividing both sides of Eq. (i) by ' $e^y$ ', we get

$$e^{-y} dy = e^x dx + x^2 e^{x^3} dx$$

Integrating,  $\int e^{-y} dy = \int e^x dx + \int x^2 e^{x^3} dx$

$$\text{or, } -e^{-y} = e^x + \int x^2 e^{x^3} \frac{d(x^3)}{3x^2}$$

$$\text{or, } -e^{-y} = e^x + \frac{1}{3} \int e^{x^3} d(x^3)$$

$$\text{or, } -e^{-y} = e^x + \frac{1}{3} e^{x^3} + c$$

Therefore, the general solution is  $-e^{-y} = e^x + \frac{1}{3} e^{x^3} + c.$

# Separation of variables

**Problem#3.** Solve the ODE:  $\frac{d y}{d x} = (4x + y + 1)^2$

Solution: Given  $\frac{d y}{d x} = (4x + y + 1)^2 \dots \dots \dots (i)$

Let  $4x + y + 1 = z$

Then,  $4 + \frac{d y}{d x} = \frac{d z}{d x}$

Therefore the given equation becomes,  $\frac{d z}{d x} - 4 = z^2$

$$\frac{dz}{z^2 + 4} = dx$$

$$\text{Integrating, } \int \frac{1}{z^2 + 2^2} dz = \int dx$$

$$\text{or, } \frac{1}{2} \tan^{-1} \left( \frac{z}{2} \right) = x + c$$

$$\text{or, } \frac{1}{2} \tan^{-1} \left( \frac{4x + y + 1}{2} \right) = x + c.$$

Therefore, the general solution is  $\frac{1}{2} \tan^{-1} \left( \frac{4x + y + 1}{2} \right) = x + c.$

# Separation of variables

## Practice Problem

1. Put the following equation in separated form. Do not integrate.

$$a) \frac{dy}{dx} = \frac{x^2y - 4y}{x + 4}$$

$$b) \frac{dy}{dx} = \sec(y)e^{x-y}(1+x)$$

$$c) \frac{dy}{dx} = \frac{xy}{(x+1)(y+1)}$$

$$d) \frac{d\theta}{dt} + \sin \theta = 0$$

Solution to the problem 1 is:

1.

$$a) \frac{1}{y} dy = \frac{x^2 - 4}{x + 4} dx$$

$$b) e^y \cos y dy = e^x(1+x) dx$$

$$c) \frac{y+1}{y} dy = \frac{x}{(x+1)} dx$$

$$d) \frac{1}{\sin \theta} d\theta = -dt$$

# Separation of variables

## Practice Problem

1. The solution of  $y' = \sqrt{xy}$  with initial value  $y(1) = 0$  is given by

a)  $y(x) = \frac{1}{9}(x^{1/2} - 1)^2$

b)  $y(x) = \frac{1}{9}(x - 1)^2$

c)  $y(x) = \frac{1}{9}(x^{3/2} - 1)^2$

d)  $y(x) = \frac{1}{9}(x^2 - 1)^2$

2. The solution of  $y^2 - xy' = 0$  with initial value  $y(1) = 1$  is given by

a)  $y(x) = \frac{1}{1 - \ln x}$

b)  $y(x) = \frac{1}{1 - 2 \ln x}$

c)  $y(x) = \frac{1}{1 + \ln x}$

d)  $y(x) = \frac{1}{1 + 2 \ln x}$

3. The solution of  $y' + (\sin x)y = 0$  with initial value  $y(\pi/2) = 1$  is given by

a)  $y(x) = e^{\sin x}$

b)  $y(x) = e^{\cos x}$

c)  $y(x) = e^{1 - \sin x}$

d)  $y(x) = e^{1 - \cos x}$

### Solutions to the Practice quiz: Separable first-order odes

1. c.  $\frac{dy}{dx} = x^{1/2}y^{1/2}$ ,  $y(1) = 0$ .

$$\int_0^y \frac{dy}{y^{1/2}} = \int_1^x x^{1/2} dx; \quad 2y^{1/2} = \frac{2}{3}(x^{3/2} - 1); \quad y = \frac{(x^{3/2} - 1)^2}{9}.$$

2. a.  $x \frac{dy}{dx} = y^2$ ,  $y(1) = 1$ .

$$\int_1^y \frac{dy}{y^2} = \int_1^x \frac{dx}{x}; \quad -\left(\frac{1}{y} - 1\right) = \ln x; \quad \frac{1}{y} = 1 - \ln x; \quad y = \frac{1}{1 - \ln x}.$$

3. b.  $\frac{dy}{dx} = -(\sin x)y$ ,  $y(\pi/2) = 1$ .

$$\int_1^y \frac{dy}{y} = -\int_{\pi/2}^x \sin x dx; \quad \ln y = \cos x; \quad y = e^{\cos x}.$$

# Separation of variables

## Practice Problem

1. Solve the ODE  $\frac{x}{y}(dx + dy) + (x + y)(y dx - x dy) = 0$ .

2. Solve the following separable first-order equations:

(i)  $\frac{dy}{dx} = 4x\sqrt{y}$ , with  $y(0) = 1$ .

(ii)  $\frac{dx}{dt} = x(1 - x)$ , with  $x(0) = x_0$  and  $0 \leq x_0 \leq 1$ .

3. Solve the ODE  $y\sqrt{1+x^2} dy - x\sqrt{1+y^2} dx = 0$ .

4. Solve the ODE  $3e^{2x} \sec^2 3y dy + 2(e^{2x} - 1) \tan 3y dx = 0$ .

5. Solve the ODE  $xy^4 dx + (y^2 + 2)e^{-3x} dy = 0$ .

# Homogeneous Equations (of degree zero)

Form of equation:  $\frac{d y}{d x} = f(x, y)$

$f(x, y)$  is a function homogeneous of degree zero, i.e.,  $f(tx, ty) = t^0 f(x, y) = f(x, y)$ .

A function homogeneous of degree  $n$  can be defined as

$$f(tx, ty) = t^n f(x, y).$$

Example: solve  $\frac{d y}{d x} = -\frac{x^2 + y^2}{2xy}$

$$\text{Let, } f(x, y) = \frac{x^2 + y^2}{2xy}$$

$$\therefore f(tx, ty) = \frac{(tx)^2 + (ty)^2}{2(tx)(ty)} = \frac{t^2(x^2 + y^2)}{t^2(2xy)} = t^0 \frac{(x^2 + y^2)}{(2xy)} = t^0 f(x, y)$$

Therefore  $f(x, y)$  is a homogeneous function of degree zero

# Homogeneous Equations (of degree zero)

Another Definition: A homogeneous equation of 1<sup>st</sup> order and 1<sup>st</sup> degree is can be written in the following form:  $\frac{dy}{dx} = f\left(\frac{y}{x}\right) \dots \dots \dots (*)$

For a solution of this equation,

$$\text{Let } \frac{y}{x} = v, \text{ i.e. } y = vx$$

$$\text{or, } \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Substituting this into (\*),

$$\text{or, } v + x \frac{dv}{dx} = f(v)$$

# Homogeneous Equations (of degree zero)

$$\text{or, } v + x \frac{dv}{dx} = f(v)$$

$$\text{or, } x \frac{dv}{dx} = f(v) - v$$

$$\text{or, } \frac{dv}{f(v) - v} = \frac{dx}{x} \quad [\text{separating variables}]$$

Solve by applying the separation of variables, then you will get a solution of the general equation of the first order and degree  $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ .

# Homogeneous Equations (of degree zero)

2. Solve:  $(x^2 + y^2) dx + 2xy dy = 0$

$$\text{or, } \frac{dy}{dx} = -\frac{x^2 + y^2}{2xy}$$

For a solution of this equation,

$$\text{Let } \frac{y}{x} = v, \text{ i.e. } y = vx$$

$$\text{or, } \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Substituting this into (1),

$$v + x \frac{dv}{dx} = -\frac{x^2 + v^2 x^2}{2x \cdot vx}$$

$$\text{or, } v + x \frac{dv}{dx} = -\frac{1 + v^2}{2v}$$

$$\text{or, } v + x \frac{dv}{dx} = -\frac{1 + v^2}{2v}$$

$$\text{or, } x \frac{dv}{dx} = -\frac{1 + v^2}{2v} - v$$

$$\text{or, } x \frac{dv}{dx} = -\frac{1 + 3v^2}{2v}$$

$$\text{or, } -\frac{2v}{1 + 3v^2} dv = \frac{1}{x} dx$$

$$\text{Integrating, } -\int \frac{2v}{1 + 3v^2} dv = \int \frac{1}{x} dx$$

$$\text{Or, } -\int \frac{2v}{1 + 3v^2} \frac{d(1 + 3v^2)}{6v} = \int \frac{1}{x} dx$$

$$\text{Or, } -\frac{1}{3} \ln(1 + 3v^2) + \ln c = \ln(x)$$

# Homogeneous Equations (of degree zero)

## Self Study

$$\text{Or, } \ln(x) + \frac{1}{3} \ln(1+3v^2) = \ln c$$

$$\text{Or, } \ln(x) + \ln(1+3v^2)^{1/3} = \ln c$$

$$\text{Or, } \ln\left\{(x)(1+3v^2)^{1/3}\right\} = \ln c$$

$$\text{Or, } x(1+3v^2)^{1/3} = c$$

$$\text{Or, } x\left(1+3\frac{y^2}{x^2}\right)^{1/3} = c, \quad \because v = \frac{y}{x}$$

1. Solve the initial value problem  $(x+y)dy - ydx = 0$ ,  $y(0)=1$

2. Solve  $(x^2+y^2)dx - 2xydy = 0$ .

3. Solve  $\frac{dy}{dx} = \frac{x-y}{x+y}$

# Linear Differential Equations; Method of Integrating Factors

- Linear DEs are friendly to be solved. We can find some smooth methods to deal with.

A first-order DE of the form  $a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \dots (1)$

is said to be **a linear equation in  $y$** .

When  $g(x) = 0$ ,

(1) is said to **be homogeneous**; otherwise, it is **nonhomogeneous**.

# Linear Differential Equations; Method of Integrating Factors

- ***Standard Form***

The standard form of a first-order LDE can be written as

$$dy/dx + P(x)y = Q(x) \quad (2)$$

- ***METHOD OF SOLUTION***

Let, left part of DE (2) will be integrable if we multiply both sides by  $\mu(x)$ , as yet undetermined; thus

$$\left[ \frac{dy}{dx} + P(x)y \right] \mu(x) = Q(x)\mu(x)$$
$$\frac{dy}{dx} \mu(x) + \mu(x)P(x)y = Q(x)\mu(x) \dots (3)$$

# Linear Differential Equations; Method of Integrating Factors

If we can make the LHS of (3)

$$\frac{dy}{dx} \mu(x) + \mu(x)P(x)y = \frac{d}{dx} [\mu(x)y] \dots (4)$$

a whole derivative, (3) can be solved easily in terms of  $\mu(x)y$ .

Equation (4) can be written as

$$\frac{dy}{dx} \mu(x) + \mu(x)P(x)y = \frac{dy}{dx} \mu(x) + y \frac{d}{dx} \mu(x) \dots (5)$$

Cancelling  $\frac{dy}{dx} \mu(x)$  on both sides of DE (5), we have

# Linear Differential Equations; Method of Integrating Factors

$$\mu(x)P(x) = \frac{d}{dx} \mu(x)$$

$$\text{or, } \frac{d\mu(x)}{\mu(x)} = P(x)dx$$

$$\text{or, } \ln \mu(x) = \int P(x)$$

$$\text{or, } \mu(x) = e^{\int P(x)dx} \dots (6), \text{ Considering ingeneral } C = 1$$

Eq.(6) is called an ***integrating factor*** for equation (2).

# First Order Linear Differential Equations

**Def:** A first order differential equation is said to be **linear** if it can be written

$$\frac{dy}{dx} + p(x)y = q(x)$$

Or,  $\frac{dx}{dy} + p(y)x = q(y)$

**Method of Solution:**

**The integrating Factor:**

$$I.F. = e^{\int p(x)dx}$$

$$e^{\int p(x)dx} \frac{dy}{dx} + p(x)y e^{\int p(x)dx} = q(x) e^{\int p(x)dx}$$

$$\frac{d}{dx} \left( y e^{\int p(x)dx} \right) = q(x) e^{\int p(x)dx}$$

Integrating w.r.to x,  $\int d \left( y e^{\int p(x)dx} \right) = \int q(x) e^{\int p(x)dx} dx$

$$y e^{\int p(x)dx} = \int q(x) e^{\int p(x)dx} dx$$

$$\therefore y(x)(I.F.) = \int q(x) \times (I.F.) dx + c$$

# First Order Linear Differential Equations

**Equation Form:**  $\frac{dy}{dx} + p(x)y = q(x)$

**I.F.**  $= e^{\int p(x)dx}$

**Solution:**  $y(x)(\text{I.F.}) = \int q(x) \times (\text{I.F.}) dx + c$

**Equation Form:**  $\frac{dx}{dy} + p(y)x = q(y)$

**I.F.**  $= e^{\int p(y)dy}$

**Solution:**  $x(y)(\text{I.F.}) = \int q(y) \times (\text{I.F.}) dy + c$

# First Order Linear Differential Equations

2. Solve:  $\frac{d y}{d x} = x - 3y$

**Solution:** Given Eq. can be written as:  $\frac{d y}{d x} + 3y = x \dots \dots (i)$

Which is a first-order **linear** differential equation

Where,  $p(x) = 3$ , and  $q(x) = x$

$$\therefore I.F. = e^{\int p(x) dx} = e^{\int 3 dx} = e^{3x}$$

Therefore, the solution of Eq. (i) is:

$$y(x)(I.F.) = \int q(x) (I.F.) dx + c$$

$$\text{or, } y.e^{3x} = \int x.e^{3x} dx + c$$

$$\text{or, } y.e^{3x} = x \frac{e^{3x}}{3} - \frac{e^{3x}}{9} + c$$

|   |     |                    |
|---|-----|--------------------|
| + | $x$ | $\frac{e^{3x}}{3}$ |
| - | 1   | $\frac{e^{3x}}{9}$ |
| + | 0   |                    |

Which is the required solution

# First Order Linear Differential Equations

## Self study

1. Solve:  $\frac{dy}{dx} + 2y = e^{-2x}$ , with  $y(0) = 3/4$ .

2. Solve:  $\frac{dy}{dx} - 2xy = x$ , with  $y(0) = 0$ .

3. Solve:  $\frac{dp}{dt} + pt = 3t$ .

4. Solve the following linear ODEs:

(i)  $\frac{dy}{dx} = x - y$ , with  $y(0) = -1$ ;

(ii)  $\frac{dy}{dx} = 2x(1 - y)$ , with  $y(0) = 0$ .

## Practice quiz: Linear first-order odes

1. The solution of  $(1 + x^2)y' + 2xy = 2x$  with initial value  $y(0) = 0$  is given by

a)  $y(x) = \frac{x}{1+x}$

b)  $y(x) = \frac{x}{1+x^2}$

c)  $y(x) = \frac{x^2}{1+x}$

d)  $y(x) = \frac{x^2}{1+x^2}$

2. The solution of  $x^2y' = 1 - 2xy$  with initial value  $y(1) = 2$  is given by

a)  $y(x) = \frac{1+x}{x}$

b)  $y(x) = \frac{1+x}{x^2}$

c)  $y(x) = \frac{1+x^2}{x}$

d)  $y(x) = \frac{1+x^2}{x^2}$

3. The solution of  $y' + \lambda y = a$  with initial value  $y(0) = 0$  and  $\lambda > 0$  is given by

a)  $y(x) = a(1 - e^{\lambda x})$

b)  $y(x) = a(1 - e^{-\lambda x})$

c)  $y(x) = \frac{a}{\lambda}(1 - e^{\lambda x})$

d)  $y(x) = \frac{a}{\lambda}(1 - e^{-\lambda x})$

# First Order Non Linear Differential Equations

**Bernoulli Equations:** Bernoulli Equations. Sometimes it is possible to solve a nonlinear equation by making a change of the dependent variable that converts it into a linear equation. The most important such equation has the form:

$$\frac{dy}{dt} + p(t)y = q(t)y^n$$

Form of Equation:  $\frac{dy}{dx} + p(x)y = q(x)y^n, n \in \mathbb{R}$ .

Method of Solution: The substitution  $z = y^{1-n}$  transform the Bernoulli's equation into a linear equation in  $z$ . This linear equation, after solving and back substituting the value of  $z$  gives the solution of the Bernoulli's equation.



(a) Bernoulli

# First Order Non Linear Differential Equations

1. Solve:  $\frac{d y}{d x} + 3y = xy^2 \quad \dots \dots \dots (i)$

**Solution:** Put  $z = y^{1-2} \Rightarrow z = \frac{1}{y}$

$$\Rightarrow y = \frac{1}{z}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{z^2} \frac{dz}{dx}$$

Equation (i) becomes,  $-\frac{1}{z^2} \frac{dz}{dx} + 3\frac{1}{z} = x \frac{1}{z^2}$

$$\text{Or, } \frac{dz}{dx} - 3z = -x \quad \dots \dots \dots (ii)$$

Eq. (ii) is a first-order **linear** differential equation

Where,  $p(x) = -3$ , and  $q(x) = -x$

$$\therefore I.F. = e^{\int p(x) dx} = e^{\int -3 dx} = e^{-3x}$$

# First Order Non Linear Differential Equations

1. **Solve:**  $\frac{d y}{d x} + 3y = xy^2 \quad \dots \dots \dots (i) \quad \therefore I.F. = e^{\int p(x) d x} = e^{\int -3 d x} = e^{-3x}$

**Solution:** Therefore, the solution of Eq. (ii) is:

$$z(I.F.) = \int q(x) (I.F.) dx + c$$

$$\text{or, } ze^{-3x} = \int -xe^{-3x} dx + c$$

$$\text{or, } ze^{-3x} = - \left( x \frac{e^{-3x}}{-3} - \frac{e^{-3x}}{9} \right) + c$$

$$\text{or, } \frac{1}{y} e^{-3x} = \frac{xe^{-3x}}{3} + \frac{e^{-3x}}{9} + c$$

$$\text{or, } \frac{1}{y} = \frac{x}{3} + \frac{1}{9} + ce^{3x} \quad \Rightarrow y = 1 / \left( \frac{x}{3} + \frac{1}{9} + ce^{3x} \right)$$

|   |   |                      |
|---|---|----------------------|
| + | x | $\frac{e^{-3x}}{-3}$ |
| - | 1 | $\frac{e^{-3x}}{9}$  |
| + | 0 |                      |

# First Order Non Linear Differential Equations

## Self study

1. Solve:  $\frac{d y}{d x} + \frac{y}{x} = (\log x) y^2$  ,  $y(1)=1$

2. Solve:  $xy - \frac{dy}{dx} = y^3 \cdot e^{-x^2}$ .

3. Solve:  $\frac{d y}{d x} + y = x y^3$

# Exact Equations & Integrating Factors

- Consider a first order ODE of the form

$$M(x, y) + N(x, y)y' = 0$$

- Suppose there is a function  $\psi$  such that

$$\psi_x(x, y) = M(x, y), \quad \psi_y(x, y) = N(x, y)$$

and such that  $\psi(x, y) = c$  defines  $y = \phi(x)$  implicitly. Then

$$M(x, y) + N(x, y)y' = \frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y} \frac{dy}{dx} = \frac{d}{dx} \psi[x, \phi(x)]$$

and hence the original ODE becomes

$$\frac{d}{dx} \psi[x, \phi(x)] = 0$$

- Thus  $\psi(x, y) = c$  defines a solution implicitly.
- In this case, the ODE is said to be **exact**.

## Exact Equations - Theorem 2.6.1

- Suppose an ODE can be written in the form

$$M(x, y) + N(x, y)y' = 0 \quad (1)$$

where the functions  $M$ ,  $N$ ,  $M_y$  and  $N_x$  are all continuous in the rectangular region  $R: (x, y) \in (\alpha, \beta) \times (\gamma, \delta)$ .

Then Eq. (1) is an **exact** differential equation iff,

$$M_y(x, y) = N_x(x, y), \quad \forall (x, y) \in R \quad (2)$$

- That is, there exists a function  $\psi$  satisfying the conditions

$$\psi_x(x, y) = M(x, y), \quad \psi_y(x, y) = N(x, y) \quad (3)$$

iff,  $M$  and  $N$  satisfy Equation (2).

# Exact Differential Equations

**Equation Form:**  $M(x, y)dx + N(x, y)dy = 0$

**Test for exactness:**  $\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x}$

**General Solution :**  $\int M(x, y)dx + \int N_{(\text{Terms free from } x)}dy = C.$

# Exact Differential Equations-Example 1

- Consider the following differential equation.

$$\frac{dy}{dx} = -\frac{x+4y}{4x-y} \Leftrightarrow (x+4y) + (4x-y)y' = 0$$

Then

$$M(x, y) = x + 4y, \quad N(x, y) = 4x - y$$

and hence

$$M_y(x, y) = 4 = N_x(x, y) \Rightarrow \text{ODE is exact}$$

- From Theorem 2.6.1,

$$\psi_x(x, y) = x + 4y, \quad \psi_y(x, y) = 4x - y$$

Thus

$$\psi(x, y) = \int \psi_x(x, y) dx = \int (x + 4y) dx = \frac{1}{2} x^2 + 4xy + C(y)$$

$$\text{It follows that } \psi_y(x, y) = 4x - y = 4x + C'(y) \Rightarrow C'(y) = -y \Rightarrow C(y) = -\frac{1}{2} y^2 + k$$

# Exact Differential Equations-Example 1

Thus

$$\psi(x, y) = \frac{1}{2}x^2 + 4xy - \frac{1}{2}y^2 + k$$

- By Theorem 2.6.1, the solution is given implicitly by

$$x^2 + 8xy - y^2 = c$$

**Therefore, the general solution is:**

$$\int M(x, y) dx + \int N_{(\text{Terms free from } x)} dy = c.$$

## Exact Differential Equations-Example 2

1. **Solve:**  $(2x^3 + 4y)dx + (4x + y - 1)dy = 0 \dots \dots \dots (i)$

**Solution:** Let  $M = 2x^3 + 4y$  and  $N = 4x + y - 1$ .

$$\therefore \frac{\partial M}{\partial y} = 4 \quad \text{and} \quad \frac{\partial N}{\partial x} = 4.$$

$$\text{Gives, } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

$\therefore$  Equation (i) is an exact equation.

**Therefore, the general solution is:**

$$\int M(x, y) dx + \int N_{\text{(Terms free from } x)} dy = c.$$

$$\text{Or, } \int (2x^3 + 4y) dx + \int (y - 1) dy = c.$$

$$\text{Or, } \frac{2x^4}{4} + 4yx + \frac{y^2}{2} - y = c. \quad \Rightarrow \frac{x^4}{2} + 4xy + \frac{y^2}{2} - y = c.$$

## Exact Differential Equations-Example 3

2. **Solve:**  $(y^3 - y^2 \sin x - x)dx + (3xy^2 + 2y \cos x)dy = 0 \dots \dots (i)$

**Solution:** Let  $M = y^3 - y^2 \sin x - x$  and  $N = 3xy^2 + 2y \cos x$ .

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(y^3 - y^2 \sin x - x) \text{ and } \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(3xy^2 + 2y \cos x)$$

$$\therefore \frac{\partial M}{\partial y} = 3y^2 - 2y \sin x \text{ and } \frac{\partial N}{\partial x} = 3y^2 - 2y \sin x.$$

$$\text{Gives, } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

$\therefore$  Equation (i) is an exact equation.

**Therefore, the general solution is:**

$$\int M(x, y)dx + \int N_{(\text{Terms free from } x)} dy = c.$$

$$\text{Or, } \int (y^3 - y^2 \sin x - x)dx + \int 0 dy = c.$$

$$\text{Or, } y^3 x + y^2 \cos x - \frac{x^2}{2} = c.$$

# Exact Differential Equations

## Self study

1. Solve:  $\cos(x + y)dx + (3y^2 + 2y + \cos(x + y))dy = 0.$

2. Solve:  $(1 + 2xy^3)dx + 3x^2y^2dy = 0$

3. Solve:  $(ye^{xy} \cos 2x - 2e^{xy} \sin 2x + 2x)dx + (xe^{xy} \cos 2x - 3)dy = 0.$

# Non Exact Differential Equations

If an equation is inexact (not exact), it is possible to transform

Such equation into an exact differential equation by a judicious multiplication.

A function  $I(x, y)$  is an *integrating factor* for an inexact equation if the equation

$$I(x, y)[M(x, y)dx + N(x, y)dy] = 0$$

is an exact equation.

The famous formulas to obtain an integrating factor are as follows:

**Rule:1** If  $\frac{1}{N(x, y)} \left[ \frac{\partial M(x, y)}{\partial y} - \frac{\partial N(x, y)}{\partial x} \right] \equiv g(x)$ , then  $I.F. = I(x, y) = e^{\int g(x) dx}$ .

**Rule:2** If  $\frac{1}{M(x, y)} \left[ \frac{\partial M(x, y)}{\partial y} - \frac{\partial N(x, y)}{\partial x} \right] \equiv f(y)$ , then  $I.F. = I(x, y) = e^{-\int f(y) dy}$ .

**Rule:3** If  $M = yf(xy)$ ,  $N = xg(xy)$ , then  $I.F. = I(x, y) = \frac{1}{xM - yN}$ .

**Rule:4** If  $Mdx + Ndy = 0$  is homogeneous then  $I.F. = I(x, y) = \frac{1}{Mx + Ny}$

# Non Exact Differential Equations

1. **Solve:**  $(x^2 + y^2 + x)dx + (xy)dy = 0 \dots \dots (i)$

**Solution:** Let  $M = x^2 + y^2 + x$  and  $N = xy$ .

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(x^2 + y^2 + x) \text{ and } \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(xy)$$

$$\therefore \frac{\partial M}{\partial y} = 2y \text{ and } \frac{\partial N}{\partial x} = y.$$

$$\text{Now, } \frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{xy} (2y - y) = \frac{1}{x} = g(x)$$

$$\therefore I.F. = e^{\int g(x)dx} = e^{\int 1/x dx} = e^{\ln x} = x$$

**Multiplying Eq. (i) by I.F.= x, we have:**

$(x^3 + xy^2 + x^2)dx + (x^2y)dy = 0 \dots \dots (ii)$ , which is an Exact D.E.

**Therefore, the general solution of Eq. (ii) is:**

$$\int M(x, y)dx + \int N_{\text{(Terms free from } x\text{)}} dy = c.$$

$$\text{Or, } \int (x^3 + xy^2 + x^2)dx + \int 0 dy = c. \quad \text{Or, } x^4/4 + x^2y^2/2 + x^3/3 = c.$$

Rule:1

# Non Exact Differential Equations

1. **Solve:**  $y^2 dy + ydx - xdy = 0 \dots \dots \dots (i)$

or,  $ydx + (y^2 - x)dy = 0$

**Solution:** Let  $M = y$  and  $N = y^2 - x \quad \therefore \frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(y)$  and  $\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(y^2 - x)$

$$\therefore \frac{\partial M}{\partial y} = 1 \text{ and } \frac{\partial N}{\partial x} = -1.$$

$$\text{Now, } \frac{1}{M} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{y} (1 - (-1)) = \frac{2}{y} = f(y)$$

$$\therefore I.F. = e^{-\int f(y)dy} = e^{-\int 2/y dy} = e^{-2 \ln y} = e^{\ln y^{-2}} = y^{-2} = \frac{1}{y^2}$$

Multiplying Eq. (i) by I.F. =  $\frac{1}{y^2}$ , we have:

$$\left( \frac{1}{y} \right) dx + \left( 1 - \frac{x}{y^2} \right) dy = 0 \dots \dots \dots (ii), \text{ which is an Exact D.E.}$$

Therefore, the general solution of Eq. (ii) is:

$$\int M(x, y) dx + \int N_{(\text{Terms free from } x)} dy = c.$$

$$\text{Or, } \int \left( \frac{1}{y} \right) dx + \int 1 dy = c. \quad \text{Or, } x/y + y = c.$$

Rule:2

# Non Exact Differential Equations

1. Solve:  $y(2xy+1)dx + x(1+2xy-x^3y^3)dy = 0 \dots \dots (i)$

Let  $M = y(2xy+1)$  and  $N = x(1+2xy-x^3y^3)$

$$\therefore I.F. = \frac{1}{xM - yN} = \frac{1}{xy(2xy+1) - yx(1+2xy-x^3y^3)} = \frac{1}{xy(2xy+1-1-2xy+x^3y^3)} = \frac{1}{x^4y^4}$$

Multiplying Eq. (i) by I.F. =  $\frac{1}{x^4y^4}$ , we have:

$$\frac{y(2xy+1)}{x^4y^4}dx + \frac{x(1+2xy-x^3y^3)}{x^4y^4}dy = 0 \dots \dots (ii), \text{ which is an Exact D.E.}$$

Therefore, the general solution of Eq. (ii) is:

$$\int M(x, y)dx + \int N_{(\text{Terms free from } x)} dy = c.$$

$$\text{Or, } \int \frac{y(2xy+1)}{x^4y^4}dx + \int -\frac{1}{y} dy = c.$$

$$\text{Or, } \int \left( \frac{2}{y^2x^3} + \frac{1}{y^3x^4} \right) dx - \int \frac{1}{y} dy = c.$$

$$\Rightarrow -\frac{1}{y^2x^2} - \frac{1}{3y^3x^3} - \ln y = c.$$

Rule:3

# Non Exact Differential Equations

1. Solve:  $(x^4 + y^4)dx - xy^3dy = 0 \dots \dots (i)$

**Solution:**

$$\therefore I.F. = \frac{1}{xM + yN} = \frac{1}{x(x^4 + y^4) + y(-xy^3)} = \frac{1}{x^5 + xy^4 - xy^4} = \frac{1}{x^5}$$

Multiplying Eq. (i) by I.F. =  $\frac{1}{x^5}$ , we have:

$$\frac{(x^4 + y^4)}{x^5} dx - \frac{xy^3 dy}{x^5} = 0 \dots \dots (ii), \text{ which is an Exact D.E.}$$

Therefore, the general solution of Eq. (ii) is:

$$\int M(x, y) dx + \int N_{(\text{Terms free from } x)} dy = c.$$

$$\text{Or, } \int \frac{(x^4 + y^4)}{x^5} dx + \int 0 dy = c.$$

$$\text{Or, } \int \left( \frac{1}{x} + \frac{y^4}{x^5} \right) dx + \int 0 dy = c. \quad \Rightarrow \ln x - \frac{y^4}{4x^4} = c.$$

Rule:4

# Non Exact Differential Equations

## Self study

1. Solve:  $(2x^3y^2 + 4x^2y + 2xy^2 + xy^4 + 2y)dx + 2(y^3 + x^2y + x)dy = 0.$

Rule:1

1. Solve:  $(2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0.$

Rule:2

1. Solve:  $(2y + 3xy^2)dx + (x + 2x^2y)dy = 0.$

Rule:3

1. Solve:  $y(y^2 - 2x^2)dx + x(2y^2 - x^2)dy = 0.$

Rule:4

Thanks a lot ...